POLAR WINDOW

Sun, Earth and terrain for CLPS missions at the Moon's south pole

Submission report for the 2026 challenge *CLPS Lunar Mission Browser*. Every figure below was measured on commit `97d7626`, 24 September 2026.

1 Problem statement

NASA's brief notes that south-pole missions face extreme lighting and communication constraints, and that existing Sun and Earth tools have "complicated interfaces". It asks for an intuitive tool that lets planners, educators and the public compare landing sites *and dates* quickly, with Sun and Earth shown against the horizon to judge power and direct-to-Earth windows.[1]

General tools fail at the pole for four reasons:

- **The Sun hugs the horizon.** The Moon's axis is tilted only 1.54°, so at the pole the Sun circles the horizon. At Connecting Ridge its centre stays between −1.61° and −0.70° all month, so metres of terrain decide the answer, and a smooth-globe horizon gets it wrong.
- **The textbook azimuth formula breaks down.** It divides by the cosine of latitude, which goes to zero exactly where the sites are.
- **Crater coordinates point at shadow.** An IAU crater coordinate is the centre; the elevation model puts Shackleton's at −2,767 m, a permanently shadowed floor.
- **Earth sinks for days.** Its elevation at Connecting Ridge swings from −7.4° to +6.0° over the month, so contact is lost for days, not minutes.

2 Our solution

Every screen draws one primitive: *time above the local horizon*. A body is "up" when its centre clears the site's own skyline, computed from real terrain. From the Sun and Earth series over a window follow four numbers, always shown together: (1) hours of sunlight (power); (2) hours of Earth contact (direct-to-Earth geometry); (3) the longest dark gap (survival); (4) the longest blackout (autonomy). The last two end missions, and a total hides them.

Missions come first. A mission is a published target plus a surface window. The home screen shows 20 CLPS deliveries on a Moon view and a programme timeline, filtered by instrument type, status and provider, and every record field carries its source. A mission's target becomes a site, and the same engine answers what the Sun and Earth do there. A window picker and a monthly calendar make dates as easy to compare as places. Fifteen terrain-derived candidate sites are the alternatives. Browser tests walk the mission screens end to end, and with no service at all the app still shows the demo path from a saved copy of the API's answers, labelled as such.

3 What makes it different

Real geometry, on the server. Terrain is LOLA at 20 m to 20 km and 80 m to 50 km, ray-cast on 1,440 bearings with curvature removed. Sun and Earth come from SPICE (DE440, `MOON_ME`, light-time and stellar aberration), projected with unit vectors that stay exact at the pole. Each skyline crossing is bisected between samples, so it is not rounded to the time step.

[1] `https://www.spaceappschallenge.org/2026/challenges/clps-lunar-mission-browser/`

Uncertainty travels with the number. Each duration carries a ± from the terrain's vertical error and the rate at which the body crosses the skyline. A stretch the window cut short is shown as a lower bound (≥). A crossing too shallow to time is called *grazing* rather than given a false ±. The API also brackets each total with full-disc and any-limb figures. The ± still leaves out one model choice (section 7).

No composite score. A score needs weights, and the weights belong to the mission, not to us. We rank on one named metric at a time. Each value becomes an interval, and a value the window cut short gets a range of places, never "best".

Sites derived from terrain. A candidate site is the most open-skied landable post (slope under 12°) on the high ground of a stated ring or disc around a named feature. The anchor and the rule are stored with it. Mission targets are the reverse case: we use them exactly as published, because moving one would describe a different mission.

Three faces, one dataset. Simple, Explain and Expert views change wording and density, not data: every face shows the same four numbers with their ± or ≥.

Agents that cannot state an unmeasured number. Four agents see the system only through six declared tools. A guard checks every number and full date in an answer against that run's tool results. After two failures, the answer is replaced by a summary built from measurements alone.

4 Results

Build. 295 backend tests (201 unit, 75 integration, 19 golden), 129 frontend unit and component tests and 18 browser tests pass. In the browser tests, axe reports no serious or critical WCAG 2.1 AA violation on any screen, in either theme or on a phone. Strict type checking and all eight import contracts are clean. Table 1 was measured on an Apple M3; "cold" counts only cache misses.

Table 1: Measured latency, median and 95th percentile.

Operation	N	Median	p95
Horizon mask, 1,440 bearings (precompute)	75	44.6 ms	46.7 ms
`/metrics`, cold / warm	30 / 30	11.2 ms / 0.8 ms	12.7 ms / 1.3 ms
`/timeline` (2,836 samples, 210 kB), cold / warm	30 / 30	15.2 ms / 4.1 ms	16.8 ms / 4.3 ms
`/calendar`, 365 days, monthly, cold	10	103 ms	105 ms
`/missions`, all 20, warm	30	4.0 ms	4.7 ms

Three contrasting sites over the reference window, 14 March 2027 plus 708.75 h (table 2): Connecting Ridge (high and open), Malapert (a real night) and Faustini (sheltered).

Table 2: The four numbers, in hours. ≥ marks a lower bound the window cut short.

Site	Sunlight	Earth contact	Longest dark	Longest blackout
Connecting Ridge, 89.44° S	574.4 ± 17.1	430.3 ± 5.3	121.6 ± 17.1	278.5 ± 5.3
Malapert massif, 85.99° S	394.2 ± 7.3	570.6 ± 7.6	≥ 287.8	138.1 ± 7.6
Faustini rim, 86.64° S	103.1 ± 24.0	278.6 ± 33.9	605.7, grazing	430.2, grazing

Figure 1 shows why Connecting Ridge is lit 81 % of the time although the Sun never rises. The site stands 1,957.5 m above the reference sphere, and its skyline sits below the Sun's path. Dates matter as much as places: over 2027 its longest monthly night runs from 0 h (January, and September to December) to 144.7 h (May). Bounds matter too. A window two weeks wider turns de Gerlache rim's ≥112.2 h into (202.0 ± 5.3) h.

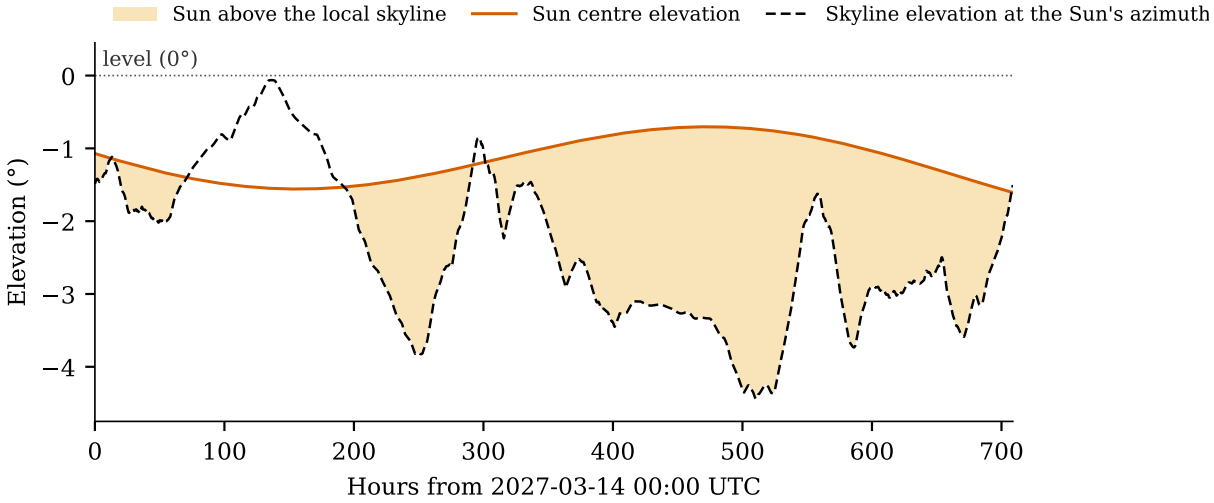


Figure 1: Connecting Ridge is lit although the Sun never rises: its centre stays below 0° all month, yet it clears the site's own lowered skyline in 81 % of samples (shaded). Measured, 900 s step.

Against flown missions. We used each mission's published LROC landing point and NASA dates. Blue Ghost 1's computed sunset (16 March 2025, 20:10 UTC) falls on the day NASA reports the mission ended, and a golden test pins it. IM-1 and IM-2 both ended while the Sun and Earth were still up. Their computed sunsets are 2 March 2024 14:01 ±28 min, with 17 % of the skyline outside the terrain model, and 19 March 2025 19:17 ±34 min. We claim no cause for either end.

CLPS data. 20 deliveries and 64 payloads, each payload with a source URL. Nine missions have a target: three LROC landing points and six IAU feature centres. None has a published landing ellipse. 139 fields are marked unconfirmed. A volatiles filter returns five missions.

5 Data sources

- **Terrain:** LOLA polar elevation models `LDEM_80S_20M` (20 m, read by HTTP range request) and `LDEM_80S_80M` (80 m, 115 MB), from the PDS Geosciences Node (LRO LOLA RDR, gridded data records).
- **Ephemeris:** NAIF kernels `de440`, `moon_pa_de440_200625`, `moon_de440_250416`, `pck00011` and `naif0012`, each pinned by SHA-256.
- **Candidate sites:** 15 anchors from the IAU Gazetteer of Planetary Nomenclature and the LPI Lunar South Pole Atlas. Four names, such as Connecting Ridge, are informal. Each point is derived by the rule in section 3.
- **Missions:** NASA CLPS delivery pages and releases, LROC images (NASA/GSFC/Arizona State University) and the Gazetteer: 48 source entries, all read on 23 September 2026. The instrument types are our own classification.

The code is released under Apache-2.0. The data are public NASA and NAIF products, credited

above. **This is not an official NASA product and is not endorsed by NASA** or by any other agency.

6 Architecture

The system is one Python service over Postgres and Redis, in layers. The `core` package is pure functions with no I/O, clock or database. Repositories alone touch SQL, services hold policy, and routes only parse and serialise. Eight import contracts enforce this. Among them: SPICE is reachable from one module only, routes never reach the ORM, and agents reach the domain only through the tool module, so the guard sees every number an agent can read.

What is precomputed follows what varies. A skyline depends only on its site, so masks and one reference window per site are built offline. Anything that depends on the window is computed per request: timelines, arcs, the four numbers, calendars and mission windows. Requests are snapped to the time grid and cached, so dragging the date reuses results.

The geometry stays on the server. The kernels (127 MB) and terrain data (308 MB) do not belong in a browser, a second copy would drift from the first, and the screens, agents and worker all call the same services. The browser keeps no geometry model: even the map's shadow line is drawn from the Sun position the API's own timeline reports. Each agent tries a chain of Gemini models, nothing older than 2.5, and moves to the next when one is over its quota.

7 Limitations and what is next

- **The 120 h survival limit is a placeholder**, standing in for a thermal and battery model. Connecting Ridge, at (121.6±17.1) h, is marked over the limit, well within its own uncertainty.
- **The terrain error is set per product, not per pixel.** Every ± rests on a placeholder vertical error (2 m or 4 m), and edge errors are assumed independent. It also leaves out the near-field noise threshold: halving or doubling it moves Connecting Ridge's longest night to 84.1 h or 206.5 h. Smooth-sphere targets have no terrain error to state, so the screens print no ± there and say why.
- **The map shows relief, not terrain shadow.** It draws the LOLA hillshade and the line where the Sun sets on a smooth Moon, from the same SPICE Sun as the numbers; ridges cast no shadow on it. The approximate illumination and PSR tiles are not drawn.
- **No screen-reader user has tested it.** Axe and 18 browser tests now run on every screen, but automated checks do not replace a person.
- **CLPS coverage is thin.** Eleven missions have no published target, and only the three flown ones have a known surface window. No mission has a published ellipse.
- **The agents have never run against a live model.** Until it is cleared, the development run log holds scripted test runs. The guard treats any whole number from 1900 to 2200 as a year, and it checks numbers, not words.
- **Two contrast gaps remain in the light theme.** The month strip's bars stand at 2.70:1 and 2.87:1 against their well, and input borders at 1.99:1, under the 3:1 a graphic needs. The fix is a colour token.

Next: validate the polar masks against published LOLA illumination maps, propagate per-pixel error, and run a live model and a screen-reader test.

8 Judge questions

Question	Answer
Science	
Why no composite score, when every competitor has one?	A score needs weights, and the weights are the mission's decision, not ours: a battery-limited lander and a rover that can hibernate rank the same sites in opposite orders. We rank on one named metric at a time. Each value becomes an interval from its own uncertainty, and a value the window cut short gets a range of places, never a rank. The trade stays visible.
How do you know the horizon masks are right?	Partly, and we say so. The ray-march is tested against surfaces whose horizon can be derived by hand (plane, cone, ramp, crater rim), and golden fixtures lock three real sites. We have not yet compared one mask with published LOLA illumination products. One near-field threshold moves Connecting Ridge's longest night from 84 h to 206 h. That comparison is the first thing to settle.
What is the error on the error?	Unknown, and larger than our error bars suggest. Every ± comes from a placeholder vertical error (2 m or 4 m by tier), with crossings treated as independent. The near-field noise threshold sits outside every ±: halving or doubling it moves Connecting Ridge's longest night by −37 h and +85 h, against a stated ±17 h. Per-pixel LOLA error and a Monte Carlo run over correlated terrain perturbations would settle it.
The Sun never rises above level at Connecting Ridge, yet you report 81 % sunlight. How?	The site stands 1,957.5 m above the reference sphere and looks down on its surroundings. Its highest skyline point is −0.06° and the mean is −2.37°. The Sun's centre stays between −1.61° and −0.70°: below level, but above that lowered skyline in 81 % of samples. "Sunlight" means above the site's own horizon, which is the definition that matters to a solar panel.
Why 0.25° bins and a 900 s step?	Near the pole the Sun's azimuth sweeps about 0.5° an hour, so one bin is roughly half an hour of Sun motion. That is close to the timing error the terrain error already implies. The step only brackets crossings, which are then bisected, so edges are not quantised to 15 minutes. A mask costs 45 ms. A convergence run at 0.125° and 300 s is still to do.
Your sunlight hours use the Sun's centre. What about the penumbra?	The API brackets the centre figure with full-disc and any-limb totals. At Connecting Ridge they are 535.6 h, 574.4 h and 603.5 h. Across the fifteen sites, 0.6 to 17.1 % of samples are partly occluded. The bracket is in the API but not yet on any screen, and it is not folded into the ±, so the headline understates this part of the uncertainty.
Why `MOON_ME`, and why light-time plus stellar aberration?	`MOON_ME` is the mean-Earth/polar-axis frame that lunar maps and the LOLA products use, so terrain and ephemeris share one frame. Light-time plus stellar aberration gives the apparent direction: where the Sun appears to a panel, where Earth appears to an antenna. We measured the correction over the window: at most 0.006° for the Sun and 0.000, 2° for Earth, far below a bin.
What does "Earth contact" mean, and not mean?	Earth's centre above the site's own skyline: the geometric possibility of a direct-to-Earth link. It does not model a link budget, antenna pattern, ground-station schedule or relay. Earth's disc is about 1.9° across, so near an edge part of it may be visible. Read it as an upper bound on when a direct link could exist, never as a communications plan.

continued overleaf

Question	Answer
Why are your site coordinates not the IAU coordinates?	An IAU crater coordinate is the centre, which near the pole is a shadowed floor: the elevation model puts Shackleton's at −2,767 m. We anchor on the named feature and search for the most open-skied landable post on its high ground. Offsets run from 0.04 to 52 km and are stored with each site. Mission targets are the exception: they are used exactly as published.
Your map shows a lit pole in March 2027; your numbers say the Sun stays below level. Which is right?	The numbers, and now the map agrees. SPICE puts the sub-solar latitude at +0.83 to +1.36° for this window, so the south pole is in winter and the Sun stays below level. Our review found the map drawing its terminator from a schematic model with the season inverted. It now takes the Sun's position from the API's own SPICE timeline and shades the side where the Sun is down on a smooth Moon.
Engineering	
What happens when SPICE kernels are missing?	The API still starts. Each of the five kernels is checked against a pinned SHA-256 and a size floor. The error names the file, its NAIF URL and the fix, and `/healthz` returns 503. Every geometry request then returns a problem detail saying what still works, never a plausible zero. Unit tests cover the messages. We have not rehearsed a demo with a kernel removed.
What happens when Redis or Postgres is down?	Redis is only a cache. Without it every request is computed, marked `bypass`, and `/healthz` reports "degraded" with a 200. Postgres holds the sites, masks and missions, so without it `/healthz` returns 503 and nothing computes. Both behaviours are designed and written down, but no automated test stops either service, and we have not rehearsed either failure.
Why is the geometry not in the browser?	One ephemeris and one terrain model give one answer. The kernels (127 MB) and elevation data (308 MB) do not belong in a browser, and a second implementation would drift from the first. The screens, the agent tools and the precompute worker all call the same services. Even the map's shadow line uses the Sun position the API reports.
What do the eight import contracts buy you?	They make the layering executable. The core has no I/O. SPICE is reachable through one module only. Routes never touch the ORM, services never import the web framework, and agents reach the domain only through the tool module, so the guard sees every number an agent could read. All eight hold on this build. They prove the structure is sound, not that the numbers are right.
What would break first at ten times the sites?	We have not measured it. Masks cost 45 ms each, so 150 sites is about seven seconds of precompute, and storage is small. The likely first limit is the single SPICE thread per process, which serialises every cold window, together with a compare call that runs sites in sequence and stops at eight. The fix is more API processes, not more threads. A 150-site load test would settle it.
Warm and cold timings, measured how?	With a script against the running API on an Apple M3, 30 requests per case. "Cold" is a window never requested before, and counts only if the response reports a cache miss. "Warm" repeats one window and requires a hit. Medians: metrics 11.2 ms cold and 0.8 ms warm; timeline 15.2 ms and 4.1 ms; a year of monthly calendar 103 ms; one mask ray-march 45 ms.

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Question	Answer
How do you know a refactor has not changed the physics?	Nineteen golden tests pin masks, timelines, the four numbers, the resolved site catalogue and one mission hindcast to committed fixtures. A manifest records the kernel digests, and regenerating the fixtures requires a stated reason. An engine version is stamped on every stored row, cache key and response. The limit is that the fixtures came from this code, so they detect change, not error.
How large is the test suite, and what does it not cover?	295 backend tests (201 unit, 75 integration, 19 golden), 129 frontend unit and component tests and 18 browser tests pass. The integration tests use their own database. The browser tests walk the demo path, check that superseded requests are cancelled, run axe on every screen in both themes and on a phone, and serve the saved copy with the service unreachable. No test stops Redis or Postgres, and there is no load test.
Product	
Why 120 h?	It is a placeholder, and the code says so: the longest darkness a lander is assumed to survive on stored energy, to be replaced by a real thermal and battery figure. The backend returns durations and never verdicts. Because the over-limit colour depends on it, Connecting Ridge at (121.6 ± 17.1) h reads as 2 h over the limit, well inside its own uncertainty.
Why should a planner trust this over an established tool?	Not yet for a commitment. It is a fast comparison layer that shows each number's source, resolution and uncertainty, and it will not invent a difference between sites. Its one external check matches Blue Ghost 1's reported end date. No polar result has yet been compared with an established illumination tool, and the error bars rest on a placeholder terrain error. Use it to shortlist and explain, then confirm with a validated tool.
Can I change the date?	Yes. A window picker sets any start and a duration from 14 days to a year. A month strip shows each month's sunlight and Earth fractions from the calendar endpoint, and a mission sets its own surface window. At Connecting Ridge the longest night ranges from 0 h in January 2027 to 145 h in May 2027, which is why dates matter as much as sites.
The challenge is a CLPS mission browser. Where are the missions?	On the home screen: 20 CLPS deliveries on a Moon view and a programme timeline, filterable by instrument type, status and provider, each opening a sourced record. The three flown missions get Sun and Earth numbers over their published surface windows. Six more have a target, and the user picks the dates. Eleven have no published target.
Is the payload list right, and what does "unconfirmed" mean?	It is compiled by hand from NASA's CLPS pages, NASA releases, LROC images and the IAU Gazetteer. Every one of the 64 payloads has a source URL, all read on 23 September 2026 and not yet re-audited. "Unconfirmed" means we found no source, so the field is left empty and listed, never guessed. There are 139 such entries, including every landing ellipse and six manifests that NASA has not fully named.
Why not move a mission's target to open sky, as you do for candidate sites?	A candidate area has no fixed point, so we choose a defensible one. A mission target is a published fact, and moving it would describe a landing that did not happen or is not planned. Where only a feature centre is published, as for the three missions aimed at Mons Mouton, the numbers are for that centre, and the record says it is not a landing point.

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Question	Answer
What about targets outside 80° S, such as Mare Crisium?	The Sun and Earth are still computed with SPICE, but against a smooth-sphere horizon. Every response flags that terrain is not modelled and notes that sunlight and Earth time are therefore upper bounds. Four of the seven target sites are like this, including Blue Ghost 1. There the screens print no ± and say the terrain is not modelled, rather than a ±0 that would read as certainty exactly where there is least.
What stops the public face being a simpler lie?	All three faces (Simple, Explain, Expert) read the same responses; only density and wording change. Simple shows the same four numbers with their ± or ≥, and its sentence about the night carries the margin or says the edges cannot be timed. What blocks Earth is read from the data at each site. Our rule is that simpler words must never mean a simpler claim.
Ethics and provenance	
What stops the AI from inventing a number?	Every number and full date in an answer is checked against that run's tool results. A failure gets one repair attempt, then the answer is replaced by a summary built only from measurements. Known holes: any whole number from 1900 to 2200 passes as a year, so an invented "2100 hours" is not caught. Words are not checked, so "best site" passes. It stops invented numbers, not invented claims.
Has the AI ever run live, and what if it is unavailable?	No. No model key has been configured, so every agent answer so far is the deterministic summary built from measurements, and it is labelled as such. The development run log still holds several hundred scripted test runs, including "ok" ones, from before the tests had their own database. It must be cleared before a demo. With a key, each agent tries a chain of Gemini models and moves on when one is over its quota. A live run is the next check.
Is this a NASA product, and what may you claim?	No. It is an independent Space Apps entry built on public NASA and NAIF data: LOLA terrain, SPICE kernels and NASA's CLPS pages. We may claim that the numbers are computed from those data. We may not claim endorsement, flight-grade values or an official list of candidate sites. The interface, the API root and the printed report each say so.
What is the most misleading thing a user could take away?	That a site under the 120 h limit is safe. The limit is a placeholder, and Connecting Ridge's night, (121.6 ± 17.1) h, lies on both sides of it. Second, that "Earth contact" is a link rather than geometry. Our review found three worse ones, now fixed: a map with the season inverted, ±0 on smooth-sphere numbers, and a sentence saying no ridge blocks Earth, false at five sites.
What did your own review find that you did not fix?	Values with an untimeable edge are ranked as if unbounded. The guard's holes. The scripted run log, until it is cleared. The placeholder terrain error and the 120 h limit. Two light-theme contrast gaps. No load test, and no test that stops Redis or Postgres. Fixed since the review: the map's season, ±0 and "uncertain by up to 0 hours" on smooth-sphere targets, the false Earth sentence, the missing ± on the simple face, stale counts in the documents, and the missing browser and accessibility tests.